

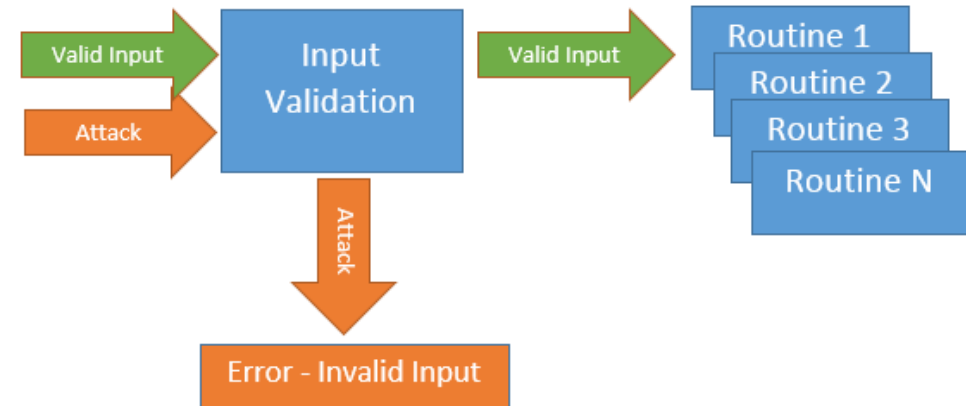
Java Full Stack Development

4. Secure Coding Practices



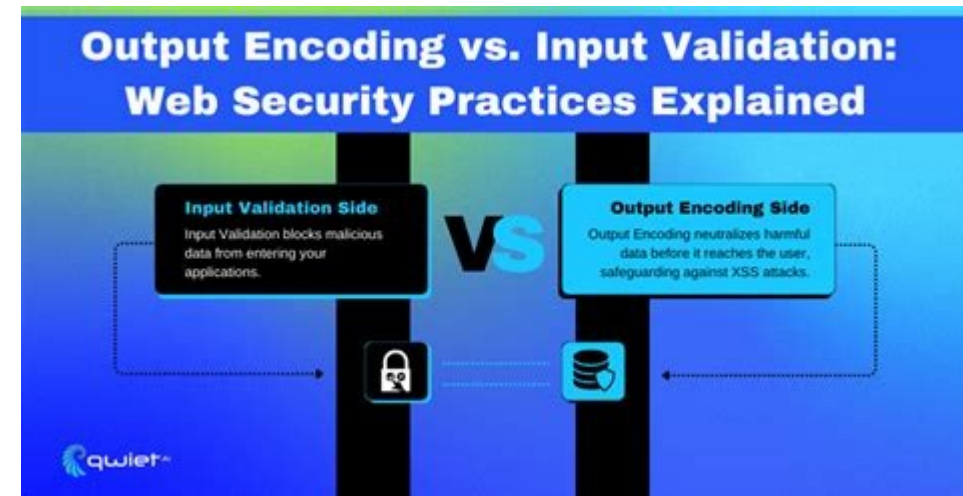
Input Validation

- Input validation
 - Validate at the boundary
 - The moment data enters your system
 - Controller, message handler, file reader
 - Validate by type, not by string content
 - Parse a date as LocalDate, not regex on a string
 - Prefer parsers over regex
 - Parsers are designed for the format
 - Regex catches some cases and misses others
 - Allowlist, not blocklist
 - Specify what's allowed, reject everything else
 - Centralize validation logic
 - One place to update when the rules change



Output Encoding

- Appropriate encoding at the output
 - Untrusted data is dangerous when interpreted by a downstream system
 - The right encoding depends on the destination context
 - HTML, JavaScript, URL, SQL, shell, log
 - Frameworks usually do this correctly by default
 - Manual configuration can break the defaults
 - Encode at the point of output, not at the point of input
 - HTML escape for HTML
 - Parameterize for SQL
 - URL-encode for URLs
 - JSON-encode for JSON



Authentication vs. Authorization

- Not synonyms
- Authentication (AuthN)
 - "Who are you?"
 - Verifying identity
- Authorization (AuthZ)
 - "What are you allowed to do?"
 - Verifying permission



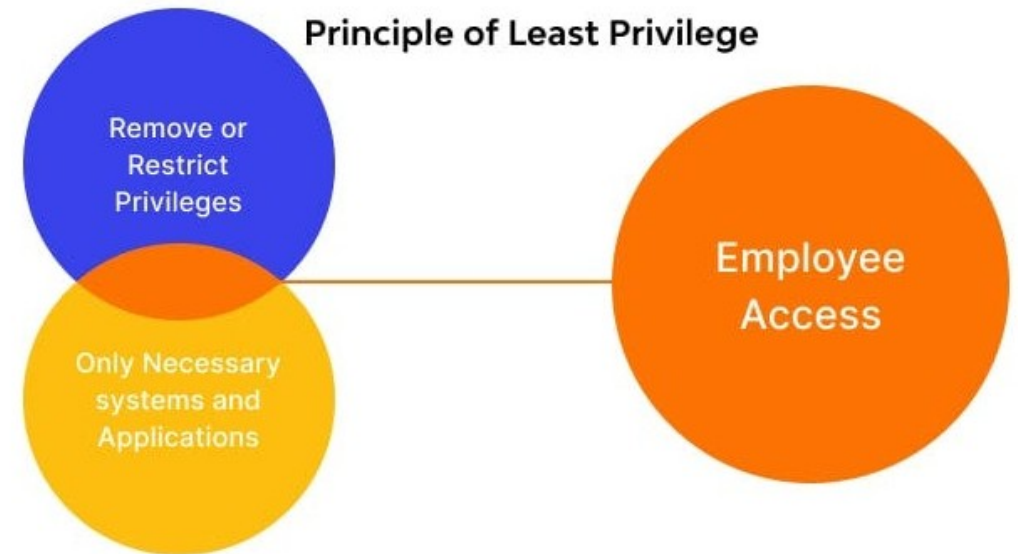
Authentication vs. Authorization

- The flow we have seen
 - Authentication produces an identity
 - Authorization decides what that identity can do
 - A failed authentication is "401 Unauthorized"
 - Badly named -- means "unauthenticated"
 - A failed authorization is "403 Forbidden"
- Confusing them produces real security bugs



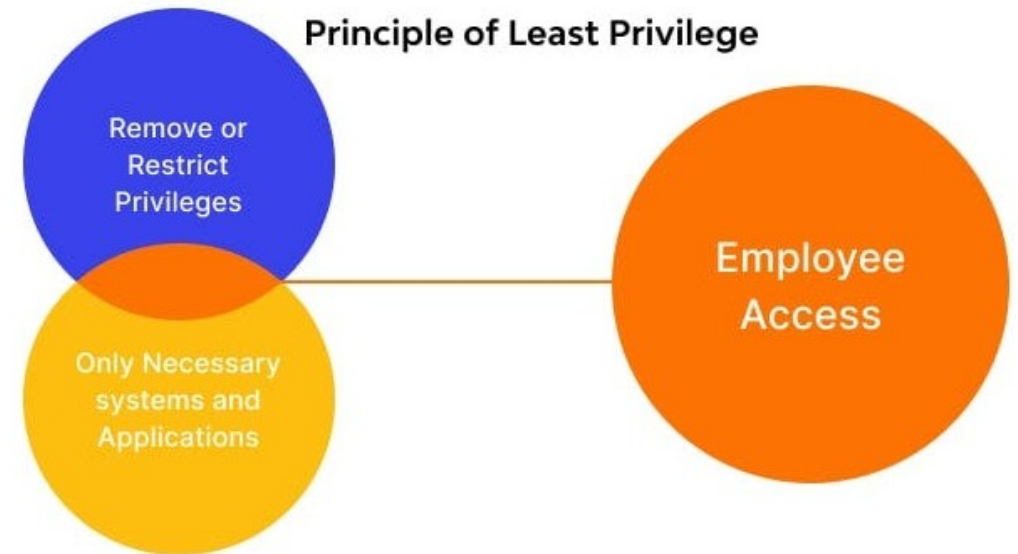
Principle of Least Privilege

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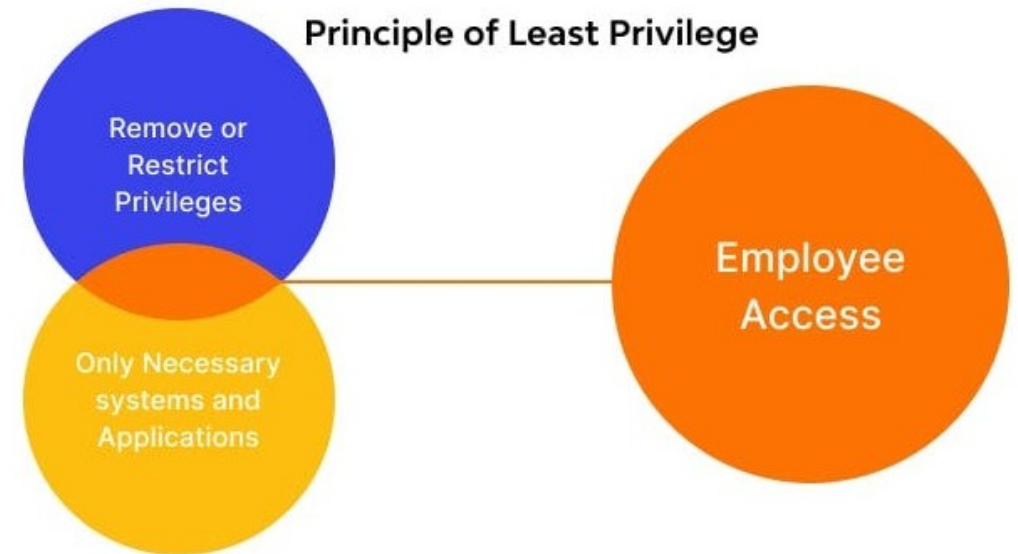
Principle of Least Privilege

- Shows up everywhere
 - OAuth scopes
 - The OAuth client requests only the scopes it needs.
 - Database accounts
 - The application's database account has only the privileges its queries need
 - A read-only reporting app should have a SELECT-only account
 - Only the actual writers like controllers that insert/update data should use a higher-privilege account, and ideally a different one



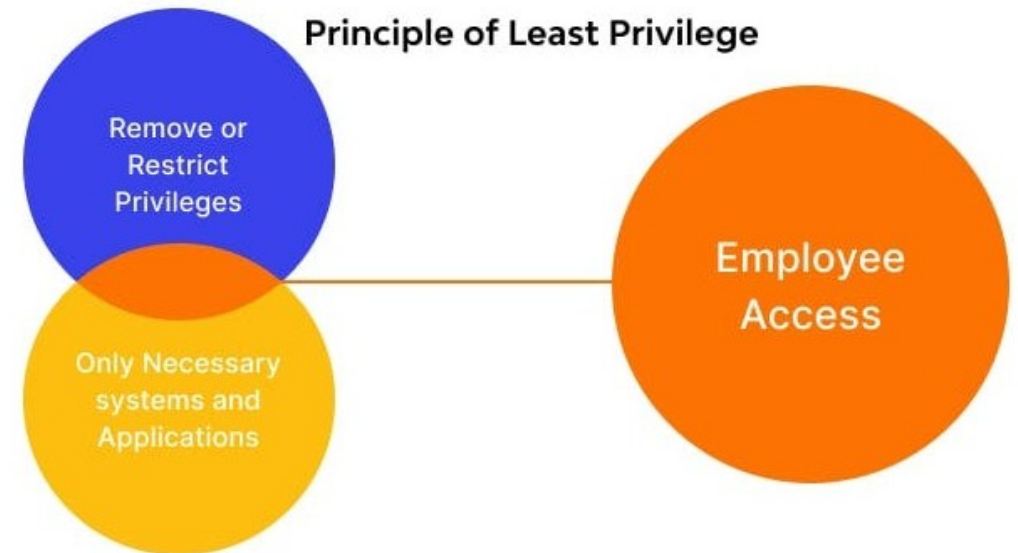
Principle of Least Privilege

- Shows up everywhere
 - Service accounts
 - When the BFF runs in production, it runs as a specific OS user
 - That user should not have admin privileges
 - It should not have write access to the application's installation directory
 - File system permissions
 - Files should be owned by users that need to read or write them, with permissions set accordingly
 - The configuration directory containing client secrets should be readable only by the service account



Principle of Least Privilege

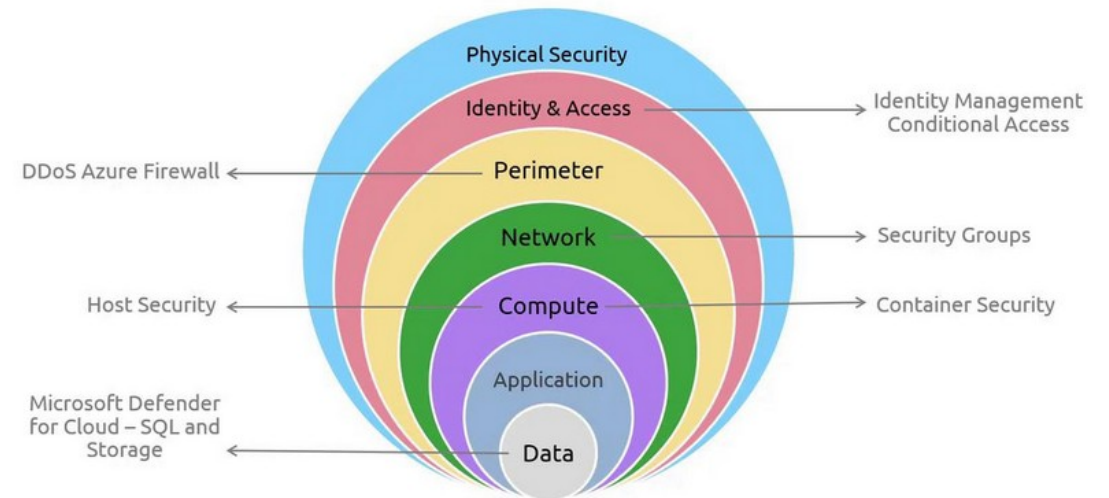
- Shows up everywhere
 - Container capabilities
 - Docker containers can be configured with reduced Linux capabilities
 - A container running a Spring Boot app should not have ADMIN or other elevated capabilities unless required.
 - Network reachability
 - The BFF needs to reach the auth server and the resource server
 - It does NOT need to reach the entire internet
 - Network policies can restrict outbound traffic to only the required hosts



Defence in Depth

- Multiple risk controls
 - Never rely on a single security control
 - Each control reduces the probability of compromise; layered controls reduce it further
 - Examples
 - Authentication: OAuth + password complexity + (production) MFA
 - Network: HTTPS + firewall + WAF
 - Application: input validation + output encoding + parameterized queries
 - Detection: logging + monitoring + alerting
 - If one control fails, the others still hold

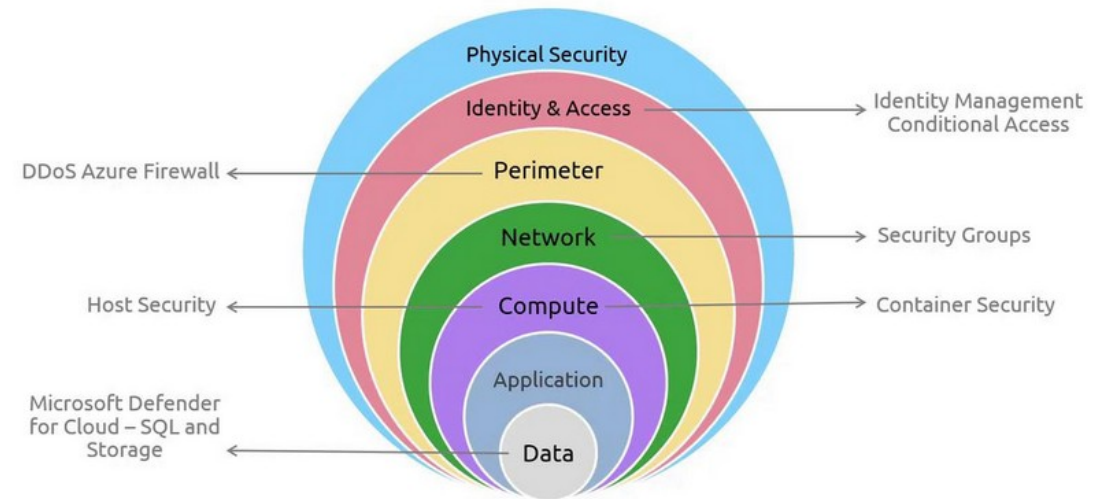
Defense in Depth



Fail Securely

- When something goes wrong
 - Default to Deny
 - The default state on error should be "deny," not "allow"
 - Common bug
 - Exception in authorization code is treated as "no rule matched"
 - So access granted
 - Correct pattern:
 - Exception in authorization
 - Log it and deny the request
 - Apply to
 - Error handlers, fallback logic, missing configuration, default value

Defense in Depth



OWASP Top Ten 2021

- Common security errors
 - A01: Broken Access Control
 - The most common category
 - Examples: a user can change a URL parameter to access another user's data
 - An admin endpoint isn't actually restricted to admins
 - The API trusts a "role" field sent by the client
 - SAST partial-detects; authorization bugs often require human review.

```
A01: Broken Access Control      <-- "users accessing things they shouldn't"
A02: Cryptographic Failures    <-- weak crypto, hardcoded keys, plaintext storage
A03: Injection                 <-- SQL, command, LDAP, XPath, etc.
A04: Insecure Design           <-- design-level flaws, not just code bugs
A05: Security Misconfiguration <-- defaults left on, debug enabled
A06: Vulnerable and Outdated Components <-- old libraries with known CVEs
A07: Identification/Auth Failures <-- weak passwords, session fixation
A08: Software and Data Integrity Failures <-- unsigned updates, untrusted deserialization
A09: Security Logging/Monitoring Failures <-- can't detect attacks you don't log
A10: Server-Side Request Forgery <-- code making requests to attacker-controlled URLs
```

OWASP Top Ten 2021

- Common security errors
 - A02: Cryptographic Failures
 - Using weak algorithms
 - Hardcoded keys
 - Plaintext password storage
 - Plaintext sensitive data at rest
 - SAST detects most of these.
 - A03: Injection
 - SQL injection
 - Command injection
 - LDAP injection
 - XPath injection
 - SAST detects most of these.

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OWASP Top Ten 2021

- Common security errors
 - A04: Insecure Design
 - Architectural flaws that aren't fixable by code-level changes
 - A feature whose design is fundamentally insecure
 - Not SAST-detectable
 - Requires threat modeling and design review.
 - A05: Security Misconfiguration
 - Default credentials still active
 - Error messages revealing too much
 - Debug mode enabled
 - Security headers missing
 - Mix of SAST and DAST detection.

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OWASP Top Ten 2021

- Common security errors
 - A06: Vulnerable and Outdated Components
 - Using a library version with known CVEs. SCA
 - Dependency scanning tools catch this
 - A07: Identification and Authentication Failures
 - Weak password policies
 - Session fixation
 - Credential stuffing not protected
 - MFA not enforced
 - Mix of SAST and design review.

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OWASP Top Ten 2021

- Common security errors
 - A08: Software and Data Integrity Failures
 - Unsigned software updates
 - Deserializing untrusted data without integrity checks
 - Dependencies from untrusted sources
 - SAST detects unsafe deserialization
 - Supply chain integrity needs additional tooling.
 - A09: Security Logging and Monitoring Failures
 - Not logging security events
 - Not alerting on suspicious patterns
 - Hard to detect with tools
 - Requires audit and process review

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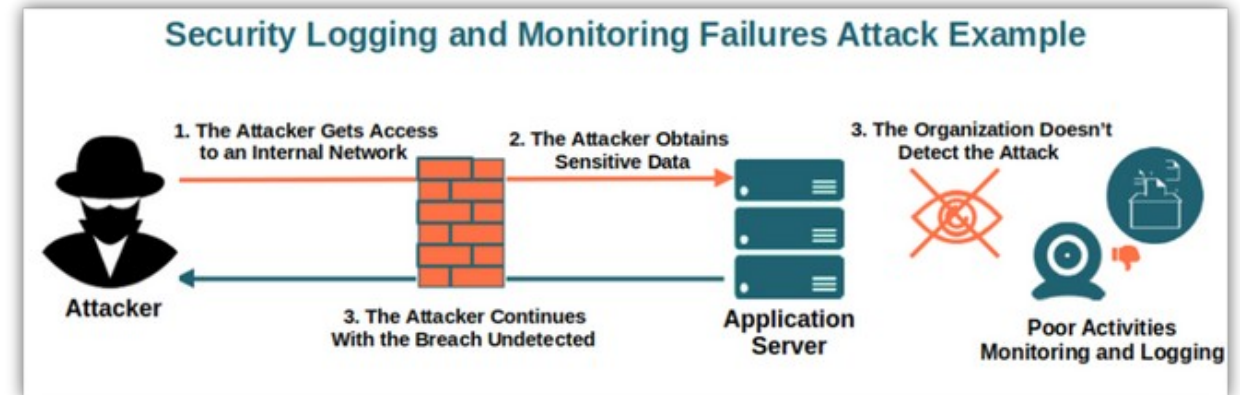
OWASP Top Ten 2021

- Common security errors
 - A10: Server-Side Request Forgery (SSRF)
 - Application making HTTP requests to URLs controlled by attackers
 - Allowing access to internal resources
 - SAST partial-detects.

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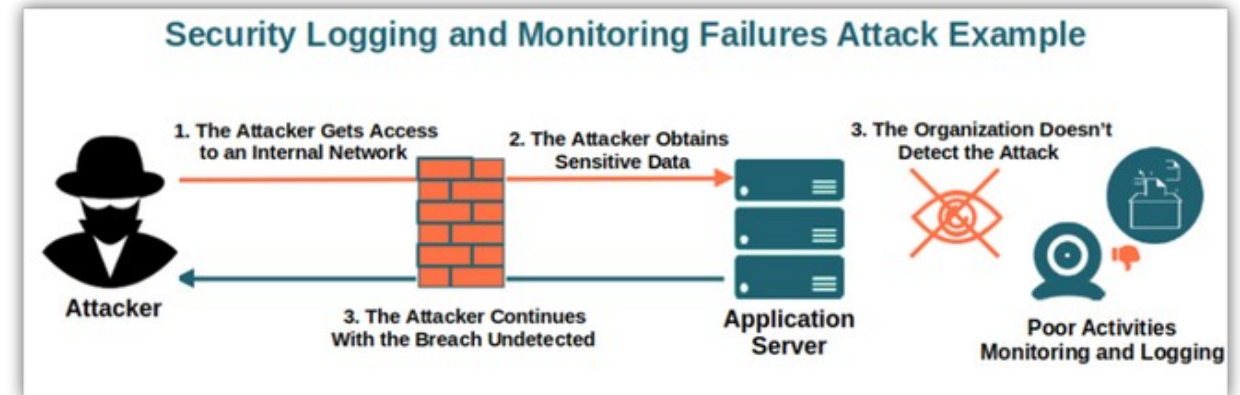
Secure Logging

- Logs: common source of breaches
 - Log security-relevant events
 - Logins, logouts, authorization failures, configuration changes, admin actions
 - Never log secrets
 - Passwords, tokens, full credit card numbers, social security numbers, API keys
 - Beware of log injection
 - Untrusted data in log messages can corrupt log files or trick log analysis tools
 - Use structured logging
 - JSON or key-value pairs, not free-form text
 - Logs are sensitive
 - Restrict who can read them, retain them per policy, encrypt at rest



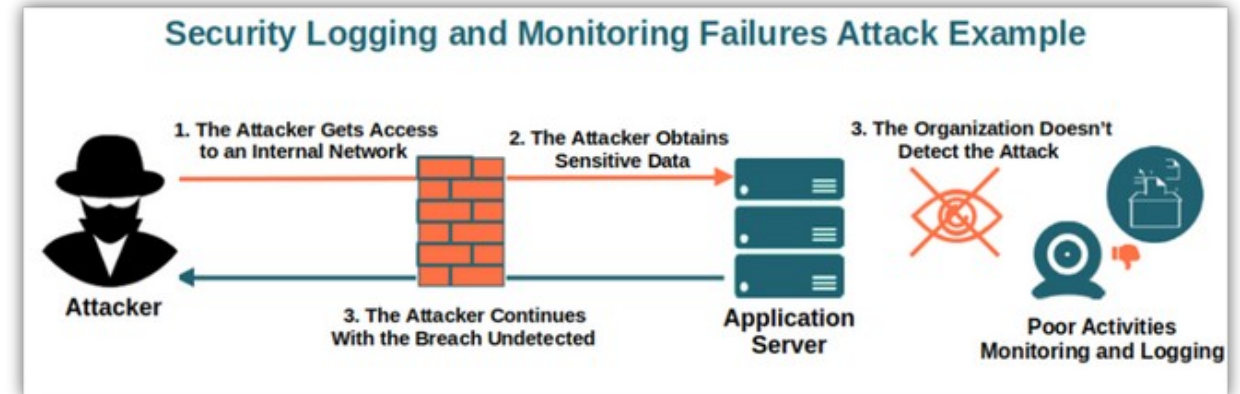
Secure Logging

- Log files are routinely
 - Stored on disk for weeks or months
 - Replicated to centralized logging systems
 - Indexed and searchable
 - Available to operations staff, support staff, and sometimes developers
 - Backed up to long-term storage
 - Sometimes shared with third-party log analysis vendors



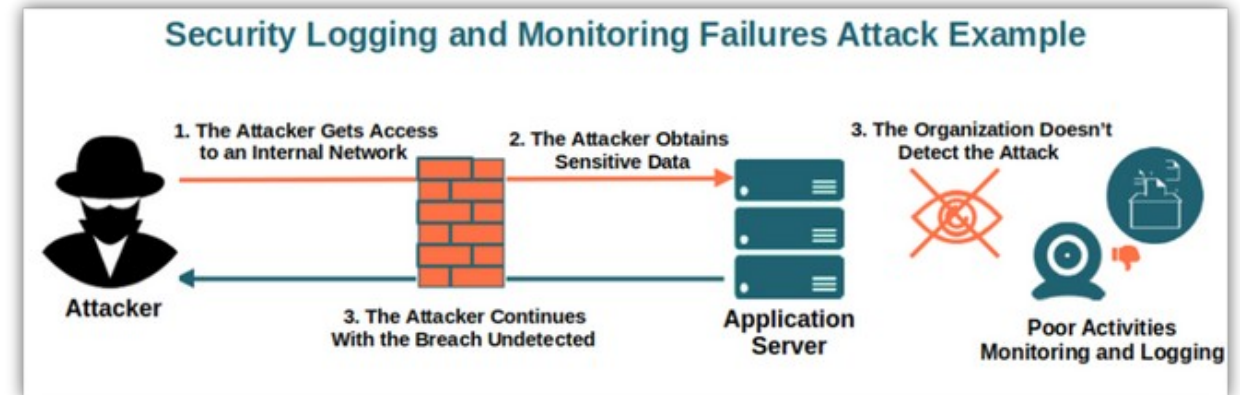
Secure Logging

- What to log
 - Authentication events
 - Login success, login failure, logout
 - Authorization decisions
 - Especially failures
 - Configuration changes
 - Administrative actions
 - Errors and exceptions
 - Performance metrics for security-relevant operations



Secure Logging

- What never to log
 - Passwords, in any form
 - Plaintext, hashed, encrypted
 - OAuth access tokens, refresh tokens, ID tokens
 - Session cookies or session IDs
 - Credit card numbers
 - Social Security numbers, government IDs
 - Private encryption keys
 - Personal health information
 - HIPAA scope
 - Anything else covered by data protection regulations



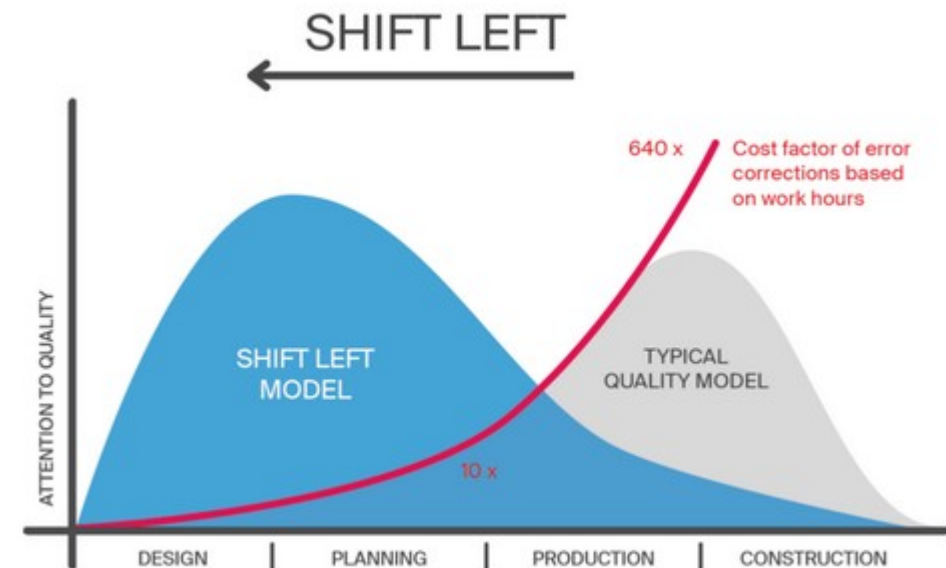
Dependency Management

- A typical Spring Boot application has
 - 100+ direct dependencies and 1000+ transitive ones
 - Each dependency is a potential vulnerability vector
 - Keep dependencies current
 - Old versions accumulate known vulnerabilities
 - Use SCA tools
 - Automated detection of vulnerable dependencies
 - Understand transitive dependencies
 - You don't directly use most of them, but they run in your process
 - Pin versions
 - Know exactly what you're shipping
 - Reproducible builds



Shift Left

- Shift-left
 - Find security issues earlier in the SDLC
 - Where they're cheaper to fix
 - IDE plugins flag issues as you type
 - SAST in CI catches what you missed
 - Pre-deploy DAST catches runtime issues



Questions?

